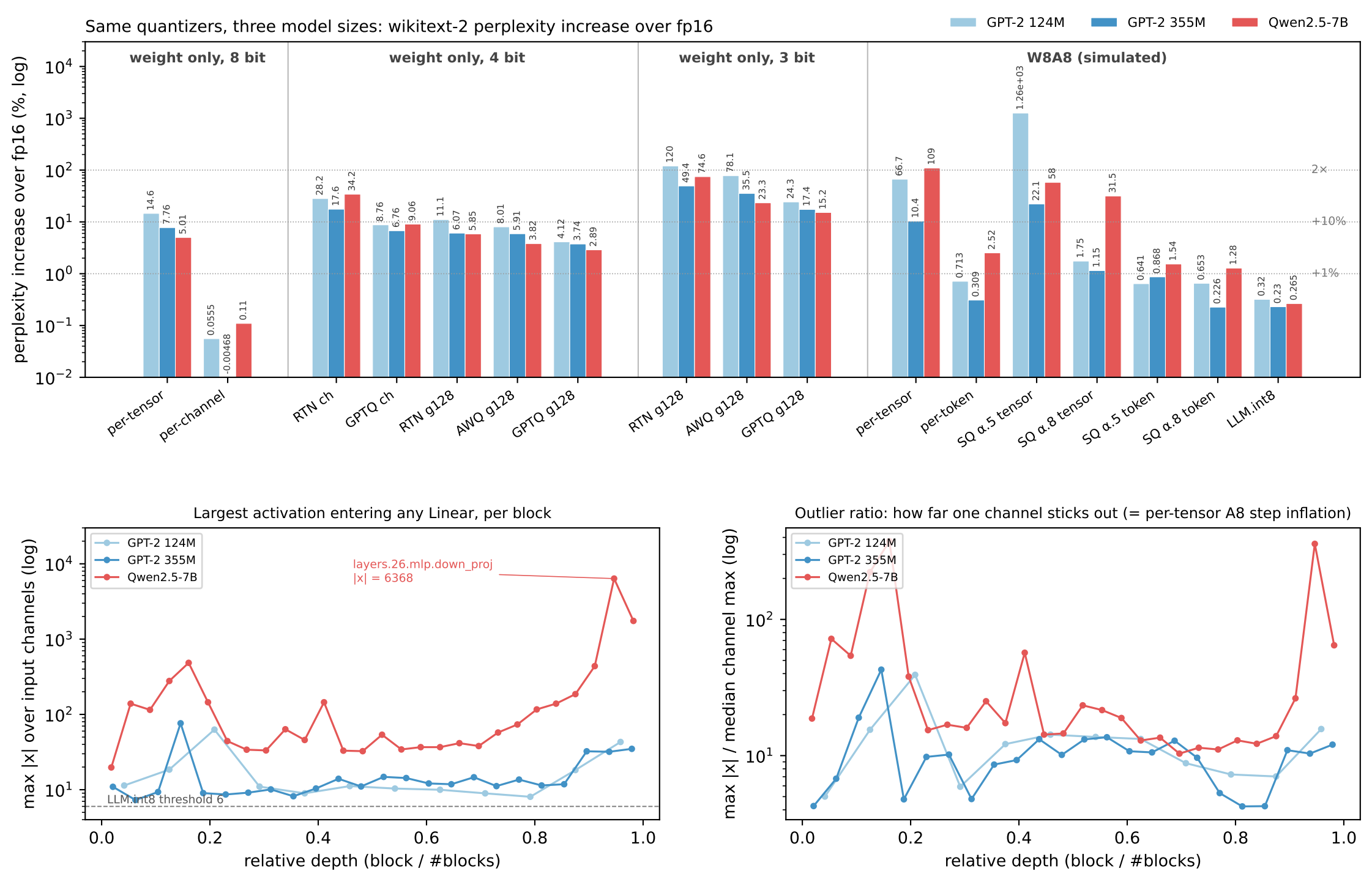
Same quantizers, three model sizes: wikitext-2 perplexity increase over fp16
GPT-2 124M
GPT-2 355M
Qwen2.5-7B
perplexity increase over fp16 (%, log)
weight only, 8 bit
weight only, 4 bit
weight only, 3 bit
W8A8 (simulated)
2×
+10%
+1%
14.6
7.76
5.01
0.0555
-0.00468
0.11
28.2
17.6
34.2
8.76
6.76
9.06
11.1
6.07
5.85
8.01
5.91
3.82
4.12
3.74
2.89
120
49.4
74.6
78.1
35.5
23.3
24.3
17.4
15.2
66.7
10.4
109
0.713
0.309
2.52
1.26e+03
22.1
58
1.75
1.15
31.5
0.641
0.868
1.54
0.653
0.226
1.28
0.32
0.23
0.265
per-tensor
per-channel
RTN ch
GPTQ ch
RTN g128
AWQ g128
GPTQ g128
RTN g128
AWQ g128
GPTQ g128
per-tensor
per-token
SQ α.5 tensor
SQ α.8 tensor
SQ α.5 token
SQ α.8 token
LLM.int8
Largest activation entering any Linear, per block
max |x| over input channels (log)
layers.26.mlp.down_proj
|x| = 6368
LLM.int8 threshold 6
relative depth (block / #blocks)
Outlier ratio: how far one channel sticks out (= per-tensor A8 step inflation)
max |x| / median channel max (log)
relative depth (block / #blocks)